TRUSTMED — Comprehensive Final Clinical AI Evaluation Report

Self-Aware Multi-Specialty Medical AI for Reliable Disease Detection
Hardware: Intel Core i5-1334U (CPU) | Storage: 5.89 GB (Budget: 7.0 GB) | Status: PASS

1. Public Dataset Acquisition & Storage Audit (Target <= 7.0 GB)

Specialty	Dataset	Disk Files	Manifest (Train/Val/Test)	Storage
Lung	NIH ChestX-ray14	9,998 images	2,517 (1,610 / 417 / 490)	2.02 GB
Brain	BraTS 2021	5,900 slices (50 subjs)	4,720 (3,304 / 708 / 708)	0.69 GB
Skin	ISIC 2019	25,331 images	3,038 (2,124 / 452 / 462)	0.15 GB
Retina	APTOS 2019	1,150 images	1,150 (805 / 172 / 173)	3.03 GB
TOTAL	**Reproducible Subsets**	**42,379 physical files**	**11,425 entries / 40 subjs**	**5.89 GB (PASS)**

2. Real Held-Out Test Evaluation Results (Zero Leakage)

Specialty	Dataset	Evaluated	Model	Primary Metric	Sensitivity	Specificity	Calibration T
Lung	NIH CXR14	490 scans	ResNet-18	Macro AUROC: 0.6552	0.0092 (th=0.50)	0.9961 (th=0.50)	T = 1.148
Brain	BraTS 2021	708 slices	U-Net	Global Dice: 0.0473	0.0624 (Global)	0.9872 (Global)	N/A
Skin	ISIC 2019	462 images	ResNet-18	Macro AUROC: 0.8922	0.4324	0.9375	T = 1.294
Retina	APTOS 2019	173 images	ResNet-18	Macro AUROC: 0.9061	0.6666	0.9244	T = 1.330

3. Lung Multi-Label Threshold Sensitivity Analysis (Prevalence: 5.2%)

Threshold	Mean Sensitivity	Mean Specificity	Overall Accuracy	Clinical Trade-off
0.50 (Default)	0.0092	0.9961	0.9449	Near-zero false alarms; severe under-calling due to 5.2% prior
0.20	0.1261	0.9458	0.9112	Conservative diagnostic threshold
0.10	0.2587	0.8662	0.8516	Balanced multi-label detection operating point
0.05	0.4225	0.7190	0.7226	Prevalence-calibrated clinical operating point
0.02 (Val Youden J)	0.7426	0.4063	0.4364	High-sensitivity rule-out screening threshold

4. Intel Core i5-1334U Real-Time Benchmark (Measured Values)

Batch Size	Total Latency	Per-Scan Latency	Throughput	RAM RSS Footprint
Batch 1 (1 scan)	416.7 ms	416.7 ms/scan	2.40 scans/s	590.2 MB
Batch 2 (2 scans)	746.6 ms	373.3 ms/scan	2.68 scans/s	590.2 MB
Batch 3 (3 scans)	1154.4 ms	384.8 ms/scan	2.60 scans/s	590.2 MB
Batch 4 (4 scans)	1559.8 ms	389.9 ms/scan	2.56 scans/s	590.2 MB

Clinical Disclaimer: AI-assisted decision-support prototype. Experiments conducted on reproducible subsets of public datasets. Not for primary diagnosis or prescription. Human specialist review required.